Four-energy complementarity can。。。 : The case of the Zhoushan Archipelago

Abstract

1 Introduction

As the global economy grows and energy demand rises, fossil fuel-based power generation can no longer sustainably meet long-term development needs. This has made the development and utilization of new energy sources a critical issue in the global energy transition. In this context, the oceans—covering about 71% of the Earth's surface—have become a major study target owing to their immense renewable energy potential [1], [2]. However, ocean energy, especially wave and wind energy, displays large seasonal and random changes. Consequently, relying on a single marine energy source is insufficient to ensure a stable and reliable power supply due to its inherent intermittency and variability [3]. To overcome this challenge, the conception of multi-energy complementary systems (MECS) was proposed [4], has proven to be an effective approach to improve energy supply stability and economic efficiency. Early studies have demonstrated the superiority of wind-solar complementary systems [5] and combinations of different renewable energy sources [6]. Although initial research mainly concentrated on land-based energy sources, interest in marine systems has grown over the past decade. Research from the Instituto Tecnológico de Canarias (ITC) has demonstrated how wind-water storage integration may considerably boost renewable energy coverage in island systems [7]. By optimizing the capacity ratio between renewable generation and energy storage, fluctuations in output can be notably reduced, along with lower storage needs and curtailment rates [8]. Additionally, as research progresses, more metrics have been developed to assess the scientific rigor and reliability of offshore multi-energy complementary systems [9],[11],[13].

Over the past few decades, various pilot projects have been launched worldwide to explore marine energy utilization. However, constrained by technological limitations, many of these early attempts ultimately failed. The causes of failure can generally be classified into three categories. The first involves technical challenges, exemplified by

Portugal's *Aguçadoura Wave Farm*, which ceased operation after only a few months in 2008 due to bearing failures and related mechanical issues. The second concerns commercial constraints, such as the *Pelamis Wave Energy* demonstration project in the UK and Portugal, which was terminated in 2010 following financial difficulties at its parent company. The third relates to permitting and regulatory issues, illustrated by the *Roosevelt Island Tidal Energy* project in the United States, which was decommissioned in 2020 upon permit expiration. These experiences provide valuable lessons for the design and assessment of future MECS demonstrations, particularly in evaluating their long-term operational viability. In such assessments, multiple aspects must be considered, including energy utilization efficiency [12], power generation performance, economic feasibility, and curtailment rates [14].

To evaluate energy utilization across different combinations in MECS, the capacity factor is commonly used, representing the ratio of actual output to the maximum theoretical capacity of generation equipment [11,13] (Stoutenburg, 2010; Khan, 2022). System stability is typically assessed via the standard deviation (SD) of the composite CF [15] (Vázquez, 2024), while heuristic algorithms, such as genetic algorithms (GA) and bi-objective ant colony optimization (BOACA), are often employed to identify optimal allocation schemes [17,18] (Sawle, 2017; Abo-Elyousr, 2018).

Despite their widespread use, these evaluation and optimization methods have limitations. Strategies focusing solely on CF or SD may compromise average power output, traditional weighting methods are subjective, and heuristic algorithms involve trade-offs between computational complexity and solution accuracy. Addressing these challenges requires an objective and adaptive decision-making framework to balance utilization efficiency (high CF) and system stability (low SD).In this context, the entropy weight method (EWM) has emerged as an effective tool to assign objective weights based on the information content and variability of indicators [19–21]. By integrating EWM into MECS evaluation, subjectivity inherent in traditional weighting is minimized, enabling a more accurate assessment of each energy source's contribution. Furthermore, EWM can be combined with multi-objective optimization techniques to support practical decision-making while maintaining scientific rigor [20].

Accordingly, this study applies EWM to quantify the proportional contribution of wind, wave, tidal, and solar energy within a marine MECS in the Zhoushan Archipelago, China. By weighting each energy source according to the temporal variability of its

capacity factor, the method achieves an optimal trade-off between utilization efficiency and system stability. Sensitivity analyses, including curtailed energy and capture price, are further conducted to evaluate economic feasibility and the potential for mitigating price self-cannibalization.

2 Materials and Methods

This chapter aims to elaborate in detail on the resource data employed in this study, the characteristics of the research area, and the core methodology for optimizing the configuration of multi-energy complementary systems. The scientific rigor and reliability of the research are grounded in precise data inputs and a rigorous modeling framework, ensuring the scientific validity of the configuration results.

2.1 Study area

The study area covers the waters surrounding the Zhoushan Archipelago, extending from latitudes 29.4°–31.0°N and longitudes 121.5°–123.5°E, as shown in 错误!未找到引用源。. Located in the East China Sea, the Zhoushan Archipelago is the largest island group in China, Comprising 1,390 islands — the land area accounts for approximately 20% of China's total island territory — and hosting a population of approximately one million [22]. The region features a complex coastal and archipelagic geomorphology, characterized by strong and persistent wind resources [23]. In addition to being influenced by the northwest Pacific winds, it also possesses enormous wave energy potential from swell activity [24]. Moreover, the tidal energy resources in the Zhoushan Archipelago are exceptionally significant, with a theoretical average power capacity reaching 5.6×10^6 kW, accounting for over half of China's total tidal energy resources. [25]. Annual sunshine duration surpasses 2,000 hours, providing abundant solar radiation. This exceptional combination of natural endowments creates highly favorable conditions for the integrated development of wind, wave, tidal, and solar energy, making the area a promising hub for offshore renewable energy utilization.

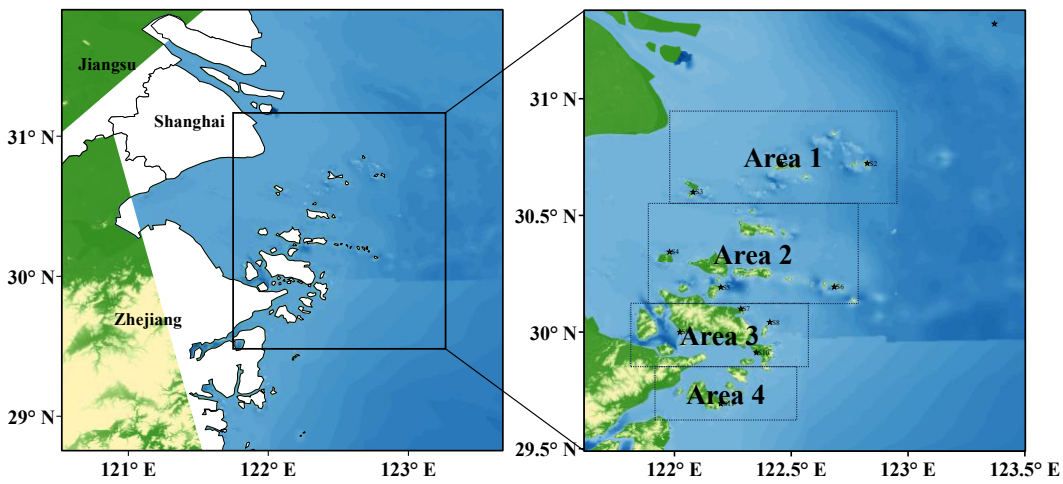

Fig. 1. Location of the study area and the high-potential development zones delineated under the pilot project.

In recent years, the Chinese government has launched a series of multi-energy complementary pilot projects in the Zhoushan Archipelago to explore the coordinated development and efficient utilization of wind energy, solar energy, tidal energy, and natural gas resources, among others. Representative projects include the megawatt-scale tidal energy demonstration project (S5) on Xiushan Island and the self-healing smart island microgrid project (S6) on Dongji Island, both of which successfully integrated tidal energy with photovoltaic power generation. These initiatives have provided continuous, stable power supply to island communities, alleviating local electricity shortages and enhancing regional energy security. Most of these pilot projects are still undergoing technological upgrades, serving as long-term experimental platforms for offshore energy integration and equipment testing.

Building upon the achievements and experiences of these pioneering projects, this study systematically identifies potential areas for multi-energy complementary development within the Zhoushan Archipelago. Although few previous studies have proposed specific spatial zoning frameworks for this purpose, this research establishes a scientific four-zone plan based on a comprehensive analysis of existing project distributions, spatial variations in energy resource endowments, and regional grid structures. Fig. 1 displays the locations of existing projects and the four designated development areas, providing a rational and practice-oriented spatial framework for optimizing future offshore multi-energy complementary systems.

To systematically guide and advance regional deployment of MECS, this study divides the Zhoushan Archipelago into four high-potential development areas (Area 1–4) based on prior pilot projects. The zoning framework primarily follows administrative

boundaries and geographic clustering, providing a clear spatial foundation for subsequent system optimization. Specific divisions are as follows:

- Area 1: Covers Shengsi County in northern Zhoushan Archipelago and its adjacent waters, characterized by unique ocean dynamics and diverse resource conditions.
- Area 2: Includes Daishan County and surrounding waters, distinguished by abundant tidal and wind energy resources.
- Area 3: Encompasses Dinghai District on Zhoushan Main Island and nearby waters. Located near the urban core, this zone requires enhanced grid stability and reliability.
- Area 4: Includes Putuo District and its surrounding waters, where energy development must consider coexistence with tourism and existing infrastructure.

This delineated zoning enables focused analysis of regional variations in energy resource availability, infrastructure conditions, and energy demand across four areas. This facilitates more efficient and cost-effective optimization of MECS and allows for the formulation of tailored development strategies for different regions based on the findings.

2.2 Data

The foundational resource assessment utilizes the ERA5 reanalysis dataset from the European Centre for Medium-Range Weather Forecasts (ECMWF), supplemented by measured wave data from the East China Sea Wave Observation Station, tidal level and current velocity data from the Zhoushan Daishan Tide Gauge Station, and topographic datasets from NOAA and GEBCO [26]. Simulation data were generated using the SWAN and MIKE numerical models and validated against field observations. Additionally, technical specifications were collected from domestic and international manufacturers of wind turbines, wave energy converters, tidal energy devices, and photovoltaic modules [27][28]. All datasets underwent model validation and standardization in accordance with national and industry standards to ensure scientific reliability and consistency.

2.3 Capacity factors

Capacity Factor (CF) is a key indicator for measuring the energy output efficiency of power generation facilities. It is defined as the ratio of the system's actual total energy output to its maximum theoretical output capacity at full load operation. In this study,

the capacity factor is used to evaluate the resource utilization potential of each renewable energy source in the Zhoushan Archipelago region. The general formula for CF is shown in Equation (1):

$$CF=\frac{\overline{P_t}}{P_e} \tag{1}$$

Among them, P_t indicates the average power, which is the ratio of actual energy output to time; P_e symbolizes the installed capacity, which is the maximum power that the power generating facility is planned to supply constantly and stably [29].

Given the distinct resource characteristics and conversion technologies of various energy sources such as wind, wave, tidal, and solar power, the specific computational models, parameter settings, and derivation processes employed for Wind power capacity factor (CF_{wind}), Wave power capacity factor (CF_{wave}), Tidal power capacity factor (CF_{tidal}), and Solar photovoltaic power capacity factor (CF_{solar}) in this research have been consolidated and detailed in Appendix A.

2.4 Weighting Allocation Methodology

Traditional weighting approaches in renewable energy allocation often rely on static or subjective assignments, which cannot adequately reflect the temporal variability and uncertainty inherent in renewable resources. To achieve a more objective and adaptive weighting mechanism, this study employs an Enhanced Entropy Weight Method (EWM). The EWM determines the weight of each indicator based on its degree of variability in the time series of capacity factors, ensuring that indicators with higher fluctuations—thus containing more information—receive greater importance in the overall evaluation.

The entropy weight method is founded on the concept of information entropy, which measures the uncertainty or dispersion of a dataset. A smaller entropy value indicates greater variability and, consequently, higher information contribution. However, when the entropy value approaches 1, even minor changes can cause exponential deviations in the computed weights, leading to unstable or unreasonable results. To address this sensitivity, the enhanced version used in this study introduces a small correction term to the normalized data, thereby stabilizing the calculation and improving robustness.

The detailed computational process is described as follows.

(1) Based on the hourly capacity factor data of multiple energy sources, the original information matrix is defined as:

$$\mathbf{X} = \left[x_{ij} \right]_{m \times n} \tag{2}$$

where x_{ij} represents the capacity factor of the j-th energy source at the i-th hour, m is the number of time samples, and n is the number of energy types.

(2) To eliminate dimensional effects and ensure comparability among indicators, the data are normalized using the min–max method:

$$r'_{ij} = \frac{x_{ij} - \min\left(x_{ij}\right)}{\max\left(x_j\right) - \min\left(x_j\right)} \tag{3}$$

When the denominator is zero, it is replaced by one to avoid division errors.
To prevent logarithmic singularities (ln 0), a small correction constant $\varepsilon = 10^{-6}$ is added:

$$r_{ij} = r'_{ij} + \varepsilon \tag{4}$$

Thus, the standardized matrix $R = r_{ij}$ is obtained.

(3) The normalized values are transformed into proportion coefficients, forming a frequency matrix:

$$f_{ij} = \frac{r_{ij}}{\sum_{i=1}^{m} r_{ij}} \tag{5}$$

The information entropy of the j-th indicator is then expressed as:

$$E_j = -k \sum_{i=1}^{m} f_{ij} \ln(f_{ij}), \quad k = \frac{1}{\ln(m)} \tag{6}$$

where $E_j \in [0,1]$.

A smaller E_j indicates that the indicator exhibits stronger variability and provides more informative contribution to the system.

(4) The divergence coefficient D_j represents the amount of useful information carried by the indicator and is defined as:

$$D_j = 1 - E_j \tag{7}$$

The entropy weight of each energy source is then calculated as:

$$\omega_j = \frac{D_j}{\sum_{j=1}^{n} D_j} \tag{8}$$

which satisfies the normalization condition $\sum_{j=1}^{n} \omega_j = 1$.
Indicators with higher temporal variability in their capacity factors receive proportionally greater weights.

2.5 Complementary analysis

Determining the optimal energy allocation ratio in MECS is a critical step toward improving overall stability and utilization efficiency. However, traditional optimization strategies exhibit several inherent drawbacks. They frequently depend on historical data to establish fixed weighting schemes, which limits adaptability to dynamic fluctuations in renewable energy resources—such as seasonal variations in wind or solar capacity factors. Moreover, subjective or empirical weight assignments often lack a rigorous quantitative foundation, reducing the scientific credibility of the resulting allocation results. In addition, the lack of flexible threshold adaptation mechanisms often leads to computational inefficiency and convergence difficulties in optimization procedures.

To address these limitations, this study integrates the enhanced entropy weight method with a complementary optimization framework. The objective is to identify an allocation scheme that achieves a balance between system stability and energy utilization efficiency.

Within conventional optimization paradigms, the Minimum Standard Deviation (MSD) approach is commonly employed, in which system stability is maximized by minimizing fluctuations in the total capacity factor (CF_t). The CF_t of the MECS at time t is defined as:

$$CF_t = \sum_{i=1}^{n} \omega_i \cdot CF_{i,t} \tag{9}$$

where $CF_{i,t}$ represents the capacity factor of the i-th energy source at time t, and ω_i is the corresponding entropy weight derived from Section 2.4.

Furthermore, the system's stability is evaluated through the SD of the CF_t, given by:

$$\sigma = \sqrt{\frac{1}{N}\sum_{t=1}^{N}(CF_t - \overline{CF_t})^2} \tag{10}$$

Where N is the number of time intervals, CF_t represents the total capacity factor at time t, and $\overline{CF_t}$ is its mean value. A smaller σ indicates higher temporal stability and stronger complementarity among energy sources.

By combining the improved entropy weighting approach with the complementary performance evaluation framework, this study provides an adaptive and objective methodology for energy allocation. The model enables dynamic response to temporal variability in renewable energy resources, ensuring both high utilization efficiency and stable system output—achieving a balanced optimization objective unattainable by

conventional single-criterion methods.

2.6 Sensitivity analysis

To evaluate the feasibility and risk resilience of various MECS configurations across the Zhoushan Archipelago, this study introduces curtailed electricity volume and capture price as sensitivity indicators. It conducts an in-depth analysis of development potential across different regions to provide more economically viable development recommendations.

Wasted electricity refers to the portion of electricity generated by the system that remains unused due to grid constraints, insufficient transmission capacity, or instantaneous output exceeding local load demand. Its calculation formula is:

$$E_{cur} = \sum_{t=1}^{8784} (\max(0, P_{gen,t} - P_{load,t}) \times \Delta t \tag{11}$$

$P_{gen,t}$ denotes the total power generation of the system at time t, calculated using the formula:

$$P_{gen,t} = \sum_{i} (CF_{i,t} \times P_{Rated,i}) \tag{12}$$

$P_{load,t}$ denotes the actual load demand or grid capacity at time t, serving as the key constraint determining whether curtailment occurs. Based on Zhang et al. (2018)[38], given that the population density of Beiji Island in China is comparable to that of the Zhoushan Archipelago, we adopt a peak load of 800 kW as the reference value. Studies by Husein et al. (2017) [39] and Bertheau et al. (2020) [40] indicate that for small islands dominated by low-energy industries (fisheries, agriculture, etc.), the average load is approximately 0.5 times the peak load. Thus, the average load for the Zhoushan Archipelago can be estimated at 400 kW. Additionally, we have adopted a dynamic load calculation method that incorporates morning and evening peak electricity consumption periods. Therefore, a sine function was selected to generate the dynamic load as an idealized representation; Δt denotes the time step, typically set to 1 hour.

The capture price represents the actual weighted average selling price of renewable energy generation. If P_{cap} exceeds the average market electricity price, it indicates that the system's output aligns well with high-price periods (such as evening peak hours), yielding high economic benefits.

$$P_{cap} = \frac{\sum_{t=1}^{8784} (P_{deliverd,t} \times Price_t \times \Delta t)}{E_{Total}} \tag{13}$$

$P_{delivered,t}$ represents the actual power delivered by the system to the grid or load at time t, calculated as follows:

$$P_{delivered,t} = P_{gen,t} - P_{cur,t} \tag{14}$$

$P_{cur,t}$ represents the curtailment power at time t.

$Price_t$ denotes the electricity market price at time t. According to the 2024 China Power Market Development Report [41], China's average market transaction price is 0.4 yuan per kilowatt-hour. E_{Sold} denotes the system's actual total annual generation.

3 Results

This study analyzes the annual variation patterns and complementary characteristics of multiple renewable energy sources within the Zhoushan Archipelago area based on the total capacity factor (CFt). CFt and its standard deviation (SD) are employed to measure energy utilization efficiency and output variability, with the former reflecting installed capacity utilization levels and the latter indicating system stability. An ideal multi-energy system should maintain a high CFt while keeping SD low, thereby achieving stable and efficient energy supply.

3.1 Temporal Characteristics of Capacity Factors

The annual variations of the four energy sources, as shown in Fig. 2, display significant seasonal differences. Wind energy peaks during winter (December to February) and declines markedly in summer. Wave energy follows a highly consistent temporal pattern with wind energy, increasing in winter alongside enhanced wind conditions. Solar energy reaches its maximum during summer (June to August), with irradiation significantly weakening in winter, demonstrating a typical inverse relationship with wind energy. Tidal energy, governed by astronomical cycles, maintains relatively stable annual output with minimal seasonal fluctuations. The quarterly box plot (Fig. 3) further indicates that wind energy exhibits the highest variability, with wave energy containing a significant number of outliers due to typhoon impacts. Solar shows some fluctuation and contains partial outliers, while tidal energy demonstrates the lowest dispersion and highest stability. Overall, the Zhoushan sea area presents a temporal pattern characterized by “simultaneous peaks in wind and wave energy, coupled with an inverse relationship in photovoltaic output,” providing a natural resource foundation for achieving multi-energy complementarity.

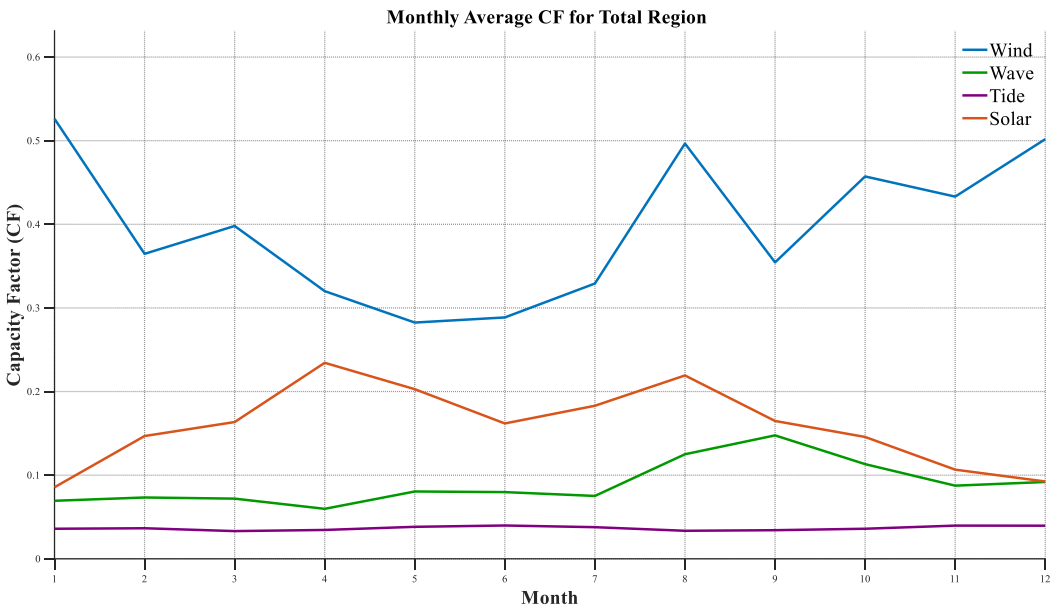

Fig. 2. Monthly CF Change Curves for Each Energy Source.

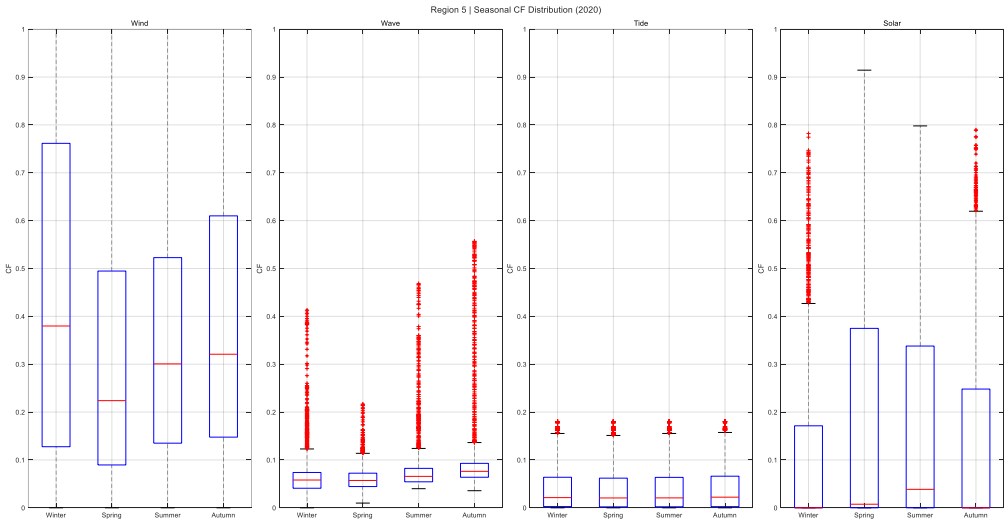

Fig. 3. Quarterly CF Box-and-Line Charts for Each Energy Source.

3.2 Temporal Complementarity Among Energy Sources

The temporal complementarity between different energy sources is quantified using correlation coefficients (Fig. 4). The correlation between wind and wave energy remains consistently high, mostly at 0.5 or above. This is primarily due to higher wind speeds in late winter and early spring, and lower speeds in late spring and early summer. From July to October, frequent typhoon activity in the northwest Pacific generates substantial swells. These swells, combined with locally induced wind waves from coastal storms and large-scale swells from offshore storms, significantly alter wave conditions in the Zhoushan Sea area. Tidal energy production is primarily influenced by celestial gravitational forces and is almost entirely independent of meteorological

conditions (wind, waves, solar radiation). Consequently, harnessing tidal energy can effectively mitigate the risks associated with energy instability. This holds significant importance for combined power generation, as it enhances system stability.

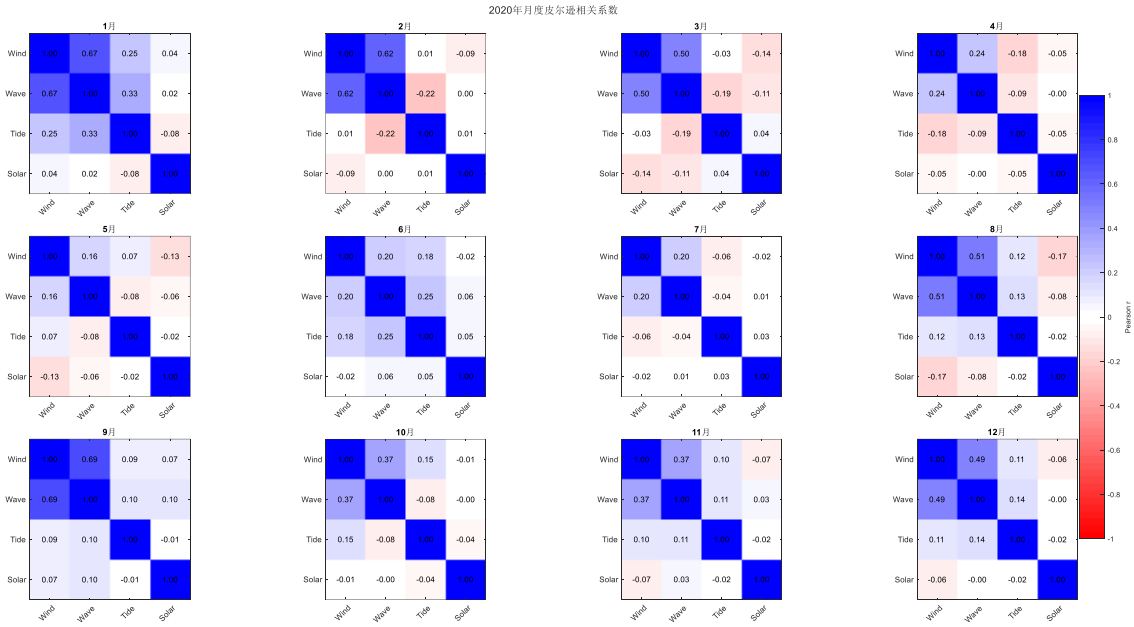

Fig. 4. Monthly Pearson Correlation Coefficients for Various Energy Sources.

3.3 Energy Allocation Weights

Based on the indicator weights determined by the entropy weight method, the energy contribution and comprehensive performance of each sub-area were further calculated (**Table 1**). Results indicate that the Zhoushan Archipelago is dominated by tidal energy (31.59%) and solar energy (44.68%), with a system-wide comprehensive CF_t of 0.1495 and a SD of 0.1114. Significant variations were observed across sub-regions: Area 1 exhibits substantial wave energy influence (62.64%), yet demonstrates poor CF_t and steady-state performance (0.0373 and 0.0357, respectively); Areas 2 and 3 rely primarily on tidal and solar energy, with CF_t values consistently ranging between 0.145 and 0.153; Area 4 exhibits a higher proportion of wave energy (34.93%) and demonstrates overall performance in the middle range.

Table 1

Calculated CF_t and σ of the MECS based on the EWM.

Area	Wind	Wave	Tide	Solar	CF_t	σ
Total	14.84%	8.90%	31.59%	44.68%	0.1495	0.1114
Area 1	9.10%	62.64%	14.56%	13.70%	0.0373	0.0357
Area 2	16.06%	8.82%	34.44%	40.68%	0.1454	0.1085
Area 3	15.90%	11.04%	33.18%	39.88%	0.1526	0.1097
Area 4	12.00%	34.93%	24.59%	28.49%	0.1188	0.0850

To compare the advantages and disadvantages between the entropy weight method and traditional approaches, we introduce energy weights calculated using the MSD method. In contrast, while the MSD method (**Table 2**) significantly reduces volatility, it tends to result in extreme weight distributions and decreased system utilization. The entropy weight method, however, achieves a balance between stability and efficiency by measuring differences in information entropy metrics, demonstrating more reasonable engineering applicability.

Table 2

Calculated CF_t and σ of the MECS based on the MSD method.

Area	Wind	Wave	Tide	Solar	CF_t	σ
Area 1	0.00171190%	99.90%	0.07%	0.03%	0.0006	0.0006
Area 2	0.00000083%	79.96%	18.13%	1.91%	0.0479	0.0327
Area 3	0.00000002%	84.20%	13.17%	2.63%	0.0464	0.0370
Area 4	0.00000603%	97.71%	1.31%	0.98%	0.0181	0.0177

The MSD method, in its pursuit of lowest volatility, exhibits an extremely concentrated weighting distribution. For instance, in Area 1, nearly all weight is allocated to wave energy, while wind, solar, and tidal energy account for negligible shares. Similarly, Area 4's mix is overly reliant on wave energy (97.71%). In contrast, the entropy-weighted method yields a more balanced allocation, particularly in rows for Area 2 and Area 3, where wave energy serves as the primary source. This aligns with the natural complementary characteristics of the Zhoushan Archipelago's overall resource endowment.

Regarding the trade-off between stability and utilization rate, the MSD method achieves the lowest std in Area 1 and Area 4, significantly lower than the entropy-weighted method. This reflects its absolute optimization for stability. However, while achieving low std, the MSD method often leads to a significant reduction in CF_t. For example, Area 1's CF_t is only 0.0006, far below the entropy-weighted method's optimized result. This confirms our earlier judgment: solely targeting minimum std may sacrifice the system's average output and energy utilization efficiency due to overly concentrated allocation.

Furthermore, **Table 1** shows that the Zhoushan Archipelago primarily targets solar and tidal energy as development sources, with wind and wave energy accounting for relatively lower shares. Although solar energy exhibits the lowest capacity factor in winter, its inherent seasonal inverse correlation with wind energy's variability confirms solar's potential as a key component in multi-energy complementary systems—

providing robust support for pilot project development. Concurrently, tidal energy's consistently high CF levels throughout the year and low variability yield advantages of low information entropy and high weighting, making it crucial for system stability. Wave energy's CF follows a seasonal pattern similar to wind energy—higher in autumn and winter, lower in summer—with significant variability, thus receiving the lowest weighting.

Based on the objective weights derived from the entropy weighting method, a configuration scheme can be established with solar and tidal energy as the main sources and wind and wave energy as supplementary sources, aiming to achieve an optimized balance between utilization rate and stability. This objective weighting method effectively avoids the subjectivity issues in weight determination inherent in traditional configuration methods.

Evaluation results for typical pilot sites (S1–S4) (**Table 3**) further validate regional differences. CF_t values at each site ranged from 0.09 to 0.11, with standard deviations below the regional average. Site S1 and S4 benefited significantly from tidal energy contributions, S3 showed dominance of wave energy, and S2 featured a higher proportion of solar power. Overall, the site-level analysis aligns with the pilot experiment's direction, providing robust scientific support for the previous pilot project.

Table 3

Calculation results of typical pilot projects based on the entropy weight method.

Area	Wind	Wave	Tide	Solar	CF_t	σ
S1	13.20%	0.00%	56.70%	30.11%	0.1035	0.0821
S2	12.06%	13.17%	44.70%	30.07%	0.1115	0.0837
S3	9.25%	49.37%	19.66%	21.72%	0.0882	0.0636
S4	9.81%	4.95%	58.03%	27.21%	0.0997	0.0724

3.4 Performance Evaluation and Regional Strategies

Further analysis combining power output and economic indicators (**Table 4**) reveals that Area 4 exhibits the highest average power and total generation capacity, yet also faces higher curtailment risks and lower capture prices. This indicates that resource development in this area requires supporting energy storage or flexible loads (such as hydrogen production and desalination) to absorb fluctuating output [44]. Area 2 and Area 3 achieve a favorable balance between stability and generation efficiency, making them suitable as priority areas for grid-connected development. While Area 1 exhibits the lowest variability, its output remains relatively low, rendering it unsuitable as a primary development zone.

Table 4

Power generation efficiency, electricity generation volume, and average power output by region.

Area	P_{Rated}(kW)	E_{Total}(MWh)	P_{Avg}(kW)
Total	1218.7	1600.3	182.19
Area 1	4105.1	1345.1	153.13
Area 2	1271	1623.3	184.8
Area 3	1391.8	1865.4	212.37
Area 4	2617.6	2732.4	311.07

3.5 Sensitivity Analysis

To further assess economic impacts, a sensitivity analysis was conducted on curtailed electricity volume and captured electricity prices (**Table 5**). Results indicate that Area 4 exhibits the highest curtailment rate (21.52%) and lowest captured electricity price (0.3186 yuan/kWh), reflecting market discounts caused by supply surplus. Area 2 demonstrates the lowest curtailment rate (6.08%) and highest captured electricity price (0.3733 yuan/kWh), highlighting its excellent load matching characteristics. After entropy weight optimization, the overall price discount rate for the system decreased from 20.35% to 17.77%. and the highest captured electricity price (0.3733 yuan/kWh), demonstrating its excellent load matching characteristics. After optimization using the entropy weight method, the overall price discount rate of the system decreased from 20.35% to 6.67%, significantly improving economic returns. This indicates that multi-energy complementary configurations not only enhance system operational stability but also effectively mitigate the price erosion effect of renewable energy in the market.

Table 5

Wasted electricity volume, wasted electricity rate, actual annual total power generation, capture cost.

Area	$E_{Curtailed}$ (MWh)	Curtailment Rate	E_{Sold} (MWh)	Pcap(Yuan/kWh)
Total	96.88	6.05%	1503.5	0.3729
Area 1	95.15	7.07%	1250.0	0.3693
Area 2	98.62	6.08%	1524.6	0.3733
Area 3	161.90	8.68%	1703.5	0.3650
Area 4	587.93	21.52%	2144.5	0.3186

In summary, the landscape resources of the Zhoushan Archipelago exhibit a typical negative correlation and complementary characteristics on a seasonal scale. The stable output of tidal energy and the winter enhancement of wave energy jointly reinforce the overall synergy of the system. The entropy weight method effectively balances

utilization rate and stability in weight allocation, demonstrating greater robustness and practical value in engineering applications compared to the MSD method.

4 Discussion

This study reveals the temporal and spatial complementary characteristics of multiple renewable energy sources—wind, wave, tidal, and solar—in the Zhoushan Archipelago region. Results indicate a significant negative correlation between wind and solar energy on seasonal scales: wind power output peaks in winter, while solar irradiance is strongest in summer. This "winter wind, summer sun" complementary pattern reflects the typical distribution characteristics of monsoon-type ocean energy, providing a natural foundation for constructing regional multi-energy complementary systems.

Fig. 2 displays that pronounced seasonal complementarity primarily stems from the annual variation patterns of the East Asian monsoon system. During winter, under the influence of the strong northwest monsoon and land-sea temperature differences, wind speeds generally increase across the Zhoushan sea area, significantly boosting wind energy output. Conversely, in summer, under the control of the subtropical high-pressure system, reduced cloud cover and improved radiation conditions lead to peak solar energy resources. The inverse correlation between these resources on seasonal scales reflects climate-driven energy conversion patterns. This cyclical variation creates a natural time-shifted relationship between wind and solar resources, effectively mitigating seasonal fluctuations in system output.

Compared with similar domestic and international studies, the findings of this research demonstrate high consistency. Lv et al. (2022) analyzed the spatiotemporal distribution and complementarity of wind and solar energy in China, identifying regions with strong complementarity primarily in the coastal areas of Zhejiang Province within the southeastern river basins [45]. Wen et al. (2025) also demonstrated a seasonally negatively correlated complementary pattern across East Asian coastal areas in their global offshore wind-solar complementarity study [46]. Our results align with these studies but reveal a more pronounced complementarity, potentially attributed to the complex island topography of the Zhoushan Archipelago and the synergistic effects of multiple energy resources. Furthermore, compared to temperate seas like the North Sea and Mediterranean, the East China Sea exhibits stronger monsoon cycles and greater seasonal amplitude in wind energy, further amplifying the negative correlation between wind and solar. This indicates that both climatic background and regional dynamic

processes jointly determine the spatiotemporal variations in multi-energy complementarity.

At the multi-energy system level, wave and tidal energy partially compensate for seasonal gaps in wind and solar output. Findings indicate that wave energy increases in tandem with wind energy during winter, while tidal energy exhibits relatively stable seasonal patterns, playing a peak-shaving and valley-filling role during spring and autumn transitions. This multi-energy coupling not only enhances system output stability but also demonstrates the synergistic potential of ocean energy within composite systems. Particularly during months with low wind-solar complementarity, tidal energy's stable output provides foundational load support, delivering significant system value.

Areas 2 & 3, with their exceptional inherent complementarity, are ideal candidates for grid-scale large-scale integration projects. In terms of configuration, maintaining a stable ratio between tidal energy and wind/solar power should be prioritized, allowing for reduced reliance on external energy storage systems and thereby lowering initial investment costs. Area 4, while possessing immense energy potential, faces high risks of curtailed electricity. Development in this area should adopt technical coupling strategies, such as integrating large-scale energy storage systems. High-capacity battery storage systems can absorb excess peak electricity and shift it for sale during high-price periods. Energy conversion can utilize curtailed electricity for water electrolysis to produce hydrogen or for seawater desalination, enabling cross-sector energy utilization that transforms low-cost electricity into high-value fuels or resources.

From an engineering and planning perspective, this complementarity holds practical significance for island-based energy systems. The wind-solar complementary characteristics in the Zhoushan region can significantly reduce the capacity requirements of energy storage systems: wind power dominates generation in winter, while solar power supplements output in summer, achieving a natural match for seasonal loads. Simultaneously, the introduction of wave and tidal energy further enhances system reliability and utilization rates. This provides a theoretical basis for promoting multi-energy coordinated supply models in island regions in the future and offers scientific reference for formulating regional energy layout plans at the policy level. The core advantage of this method lies in its data-driven and objectively weighted nature, making it applicable to any energy system facing multi-source allocation and volatility challenges. For instance, Gasa Island in South Korea or remote fishing islands

in Europe, which similarly rely on multiple marine energy sources, exhibit vastly different temporal characteristics across these resources. By inputting local resource capacity coefficient data, the model can rapidly generate objective and efficient initial allocation schemes.

It should be noted that this study's energy complementarity assessment relies on ERA5 reanalysis data, whose spatial and temporal resolutions may influence results. Additionally, the model does not yet account for the combined effects of energy storage, transmission constraints, and economic indicators, potentially introducing biases in system benefits. Future research will integrate high-resolution observational data with multi-scenario optimization models to further explore the dynamic operational characteristics and economic viability of multi-energy systems.

5 Conclusion

This study established a comprehensive evaluation system for the complementarity of wind, wave, tidal, and solar energy in the Zhoushan Archipelago region based on ERA5 reanalysis data. Using the entropy weighting method to determine the weights of each complementarity indicator, it quantified the complementary characteristics of multiple marine energy sources across temporal and spatial scales, providing methodological and theoretical support for the coordinated development of offshore renewable energy systems. Key findings are as follows:

Wind and solar energy exhibit a significant negative correlation at seasonal scales in the Zhoushan Archipelago region: wind energy peaks in winter, while solar energy is strongest in summer, demonstrating a typical monsoon-type complementary pattern characterized by “winter wind and summer sun.” This temporal complementarity significantly reduces seasonal fluctuations in system output.

Wave and tidal energy play crucial balancing roles within the multi-energy system. Wave energy increases in winter alongside enhanced wind fields, while tidal energy remains relatively stable, providing peak shaving and valley filling effects during transitional seasons. The synergistic interaction of these four energy sources significantly enhances the system's overall stability, validating the feasibility and advantages of the offshore multi-energy complementary model.

Entropy weighting analysis indicates that system volatility metrics carry the highest weight, confirming stability as the critical performance factor in multi-energy systems. Integrated complementarity index results further reveal that waters surrounding northern and central Zhoushan islands exhibit the highest multi-energy

complementarity potential, making them ideal priority zones for offshore integrated energy systems.

These findings provide valuable reference for engineering planning of island-based energy systems. By rationally configuring wind and solar capacity ratios and integrating wave and tidal energy, system reliability and renewable energy utilization can be enhanced without significantly increasing energy storage capacity. This offers scientific basis for advancing the “multi-energy complementary, source-grid-storage coordinated” energy transition on eastern coastal islands in China.

It should be noted that this study did not account for practical operational conditions such as energy storage scheduling, economic constraints, or grid integration. The results primarily reflect the complementary potential at the natural resource level. Future work will incorporate energy storage system simulations, cost-benefit analyses, and multi-scenario optimization models to refine the full-lifecycle assessment framework for offshore multi-energy systems, thereby expanding research from resource complementarity to operational optimization.

6 Acknowledgements

Appendix A. Calculation Methods for Capacity Factors

Below are the calculation models, parameter settings, and derivation processes for the capacity factors of four energy sources.

A.1. Wind power capacity factor

The wind energy capacity factor can be calculated by following the approach of Wen et al. [30]: first, average the instantaneous power Pt over a period of time (e.g., one year), then divide this average by the theoretical annual energy production of the wind turbine operating continuously at its rated power Pe to obtain the capacity factor.

$$P_t = \begin{cases} 0 & V < Cut\text{-}in\ speed \\ \frac{1}{2}\rho \cdot C_P \cdot A_T \cdot V^3 & Cut\text{-}in\ speed \le V < Rated\ speed \\ P_e & Rated\ speed \le V \le Cut\text{-}out\ speed \\ 0 & V > Cut\text{-}out\ speed \end{cases} \tag{15}$$

$$CF_{wind} = \frac{P_t}{P_e \times 8760} \tag{16}$$

In equation (2.2), ρ represents the air density, taken as 1.225 kg/m³ in this study; Cp denotes the wind energy utilization coefficient, typically set at 0.4 for modern wind turbines; AT is the swept area of the turbine blades, with a blade diameter of 154 m used in this study; V represents the actual wind speed at the turbine hub height, set at 100 m in this study.

A.2. Wave power capacity factor

The wave energy capacity coefficient can be calculated following the approach by Bahareh Kamranzad et al. [31], using the ratio of the actual annual/monthly power generation of the WEC at a specific site to its theoretical maximum energy output, as shown in Equation (2.4):

$$CF_{wave} = \frac{\sum_{i=1}^{n_T}\sum_{j=1}^{n_H} p_{ij}P_{ij}}{P_e \times 8760} \tag{17}$$

A.3. Tidal power capacity factor

As in Equation (2.5) [32], referencing the fan power curve model, replace “wind speed” with “current flow velocity (v)” and “air density” with “seawater density (ρwater = 1025 kg/m³)”.

$$P_t = \begin{cases} 0 & V < Cut\text{-}in\ velocity \\ \frac{1}{2}\rho_{water} \cdot C_P \cdot A_T \cdot V^3 & Cut\text{-}in\ velocity \le V < Rated\ velocity \\ P_e & Rated\ velocity \le V \le Cut\text{-}out\ velocity \\ 0 & V > Cut\text{-}out\ velocity \end{cases} \tag{18}$$

$$CF_{tidal} = \frac{P_t}{P_e \times 8760} \tag{19}$$

In the formula, AT represents the swept area of the tidal turbine; Ptidal denotes the rated power of the tidal energy converter; Cp is the power coefficient of the turbine (derived from rated parameters, following the same logic as wind energy).

A.4.Solar photovoltaic power capacity factor

Solar CF requires integrating the conversion efficiency of photovoltaic panels with actual solar radiation intensity (accounting for temperature and wind speed effects on panel performance). First, calculate the actual average power generation capacity, then compare it with the rated power of the photovoltaic panels. Introduce the "performance ratio PR" to correct for environmental factors. As shown in Equation (2.7) [33].

$$CF_{solar} = P_R \frac{RSDS}{RSDS_{STC}} \tag{20}$$

In equation (2.7), RSDS denotes solar radiation density, and STC represents standard test conditions (RSDSSTC = 1000 W/m²).

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